

EXERCISE

CONSIDER A ZERO-MEAN STOCHASTIC PROCESS, DEFINED BY THE FOLLOWING DATA:

$$\gamma_y(0) = 3 \quad \gamma_y(1) = 1 \quad \gamma_y(z \geq 2) = 0$$

⇒ IDENTIFY THE PARAMETERS OF THE MODEL CLASS $MA(1)$

$$y(t+1) = e(t) + \theta e(t-1) \quad e(t) \sim WN(0, \lambda^2)$$

FIRST, THE MODEL IS EXPRESSED IN T.F. FORM:

$$W(z) = \frac{1 + \theta z^{-1}}{1}$$

- SAME ORDER
- MONIC
- COPRIME
- ROOTS NOT ON UNIT CIRCLE (TO BE VERIFIED)

THIS MODEL CLASS IS SUITABLE FOR THE PROCESS, WHICH HAS $\gamma_y(z \geq 2) = 0$

1. PREDICTOR

$$\begin{aligned} \hat{y}(t+1|t) &= \frac{C(z) - A(z)}{C(z)} y(t+1) = \frac{\theta z^{-1}}{1 + \theta z^{-1}} y(t+1) \\ &= \frac{\theta}{1 + \theta z^{-1}} y(t) \Rightarrow \boxed{\hat{y}(t+1|t) = -\theta \hat{y}(t|t-1) + \theta y(t)} \end{aligned}$$

2. WRITE $\hat{J}$ AS FUNCTION OF γ_y, θ

$$\hat{J} = E[(y(t+1) - \hat{y}(t+1|t))^2] = E[\varepsilon^2(t+1)] = \gamma_\varepsilon(0)$$

VARIANCE OF 1-STEP AHEAD PREDICTION ERROR

$$\begin{aligned} &= E[y(t+1)^2] + E[\hat{y}^2(t+1|t)] - 2E[y(t+1) \cdot \hat{y}(t+1|t)] \end{aligned}$$

$$E[\hat{y}^2(t+1|t)] = E[\theta^2 (-\hat{y}(t|t-1) + y(t))^2]$$

$$\begin{aligned} \overset{\substack{\text{PREDICTOR} \\ \text{VARIANCES}}}{\gamma_{\hat{y}}(0)} &= \theta^2 \left[\overset{\substack{\text{PREDICTOR} \\ \text{VARIANCES}}}{E[\hat{y}^2(t|t-1)]} + \overset{\substack{\text{PROCESS} \\ \text{VARIANCES}}}{E[y(t)^2]} - 2E[\hat{y}(t|t-1) \cdot y(t)] \right] \\ &= \theta^2 \gamma_{\hat{y}}(0) + \theta^2 \gamma_y(0) - 2\theta^2 E[\hat{y}(t|t-1) \cdot y(t)] \end{aligned}$$

$$E[\hat{y}(t|t-1) \cdot y(t)] = E[(-\theta \hat{y}(t-1|t-2) + \theta y(t-1)) \cdot y(t)]$$

$$\begin{aligned} &\stackrel{\substack{\uparrow \\ f(y(t-1), \dots)}}{=} E[-\theta \hat{y}(t-1|t-2) \cdot y(t)] + \theta \gamma_y(1) \end{aligned}$$

$f(y(t-2), \dots) \Rightarrow \gamma_y(z \geq 2)$ WOULD APPEAR, BUT $\gamma_y(z \geq 2) = 0$!

$$= 0 + \theta \gamma_y(1)$$

so $E[\hat{y}^2(t+1|t)]$ IS SAME TO:

$$\gamma_{\hat{y}}(0) = \theta^2 \gamma_{\hat{y}}(0) + \theta^2 \gamma_y(0) - 2\theta^3 \gamma_y(1)$$

$$\gamma_{\hat{y}}(0) = \frac{1}{1 - \theta^2} [\theta^2 \gamma_y(0) - 2\theta^3 \gamma_y(1)]$$

$$\begin{aligned} E[y(t+1) \cdot \hat{y}(t+1|t)] &= E[y(t+1) (-\theta \hat{y}(t|t-1) + \theta y(t))] \\ &= \theta x_y(1) + 0 \end{aligned}$$

EVENTUALLY THE P.D.M. COST FUNCTION IS:

$$\begin{aligned} J &= x_y(0) + \frac{1}{1-\theta^2} [\theta^2 x_y(0) - 2\theta^3 x_y(1)] - 2\theta x_y(1) \\ &= \frac{x_y(0) - \cancel{x_y(0)\theta^2} + \cancel{\theta^2 x_y(0)} - 2\cancel{\theta^3 x_y(1)} - 2\theta x_y(1) + 2\cancel{\theta^3 x_y(1)}}{1-\theta^2} \\ J(\theta) &= \frac{x_y(0) - 2\theta x_y(1)}{1-\theta^2} \\ J(\theta) &= \frac{3 - 2\theta}{1-\theta^2} \end{aligned}$$

3. MINIMIZATION OF J

$$\begin{aligned} \frac{\partial J}{\partial \theta} = 0 &\Rightarrow \frac{-2}{(1-\theta^2)} + \frac{2\theta(3-2\theta)}{(1-\theta^2)^2} = 0 \\ &\Rightarrow \frac{-2(1-\theta^2) + 6\theta - 4\theta^2}{(1-\theta^2)^2} = 0 \\ \frac{-2\theta^2 + 6\theta - 2}{(1-\theta^2)^2} = 0 &\Rightarrow \theta^2 - 3\theta + 1 = 0 \end{aligned}$$

$$\begin{cases} \theta_1 = \frac{3 + \sqrt{5}}{2} \quad \times \\ \theta_2 = \frac{3 - \sqrt{5}}{2} \quad \checkmark \end{cases}$$

θ_1 IS NOT ACCEPTABLE, AS $W(z)$ WOULDN'T BE IN CANONICAL FORM AND THE CORRESPONDING PROBABLY WOULD BE UNSTABLE.

THUS θ_2 IS THE ONLY VALID SOLUTION.

WE CAN COMPUTE THE NOISE VARIANCE λ^2 :

$$\lambda^2 = \text{Var}[e(t|t-1)] = \text{Var}[e(t)] = J(\theta_2) \approx 2.61$$

THUS, THE IDENTIFIED MODEL IS:

$$M: y(t+1) = e(t) + \frac{3-\sqrt{5}}{2} e(t-1) \quad e(t) \sim WN(0, 2.61)$$